



Does it take *three* to dance the Tango? Organizational design, triadic structures and boundary spanning across subunits

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ABSTRACT

In this paper, we investigate the processes of boundary spanning across subunits within organizational networks. We hypothesize that patterns of advice across organizational subunits are explained by different triadic mechanisms depending on the organizational design of the intra-organizational network. In organizational networks characterized by limited hierarchical differentiation among members and horizontal coordination, we found triadic cyclic closure to be positively associated to boundary spanning across subunits; but when the network reflects an organizational structure with formal hierarchical differentiation among members, then we found triadic transitive closure to be associated to boundary spanning across subunits. We test these predictions in two empirical studies consisting of two organizational networks that differ from each other in organizational design. We adopt a Bayesian approach for exponential random graph models specifically designed for the statistical analysis of interpersonal relationships within and across subgroups or subunits. We suggest that the effects of informal, triadic network configurations on boundary spanning are contingent on formal organizational design.

Boundary spanning across organizational subunits is relevant for organizational processes and outcomes, including problem solving, knowledge recombination, development of new ideas and generation of innovation (Argote et al., 2003). The importance of interconnection among subunits in fostering organizational functioning features studies of organizations since the beginnings of the discipline. In his *Administrative Behavior*, Herbert Simon (1947:24) claimed, “A major task in organizing is to determine, first, where the knowledge is located that can provide the various kinds of factual premises that decisions require,” and examined whether and how organizations can mobilize coordination and the transfer of knowledge and other resources, including social support, advice and information, among their subunits. In their foundational book *Organizations*, March and Simon (1958) argued that efficient organizations must be “highly programmed with respect to the specific steps needed to accomplish the task” (page 14), including inter-personal and inter-unit coordination as a way to “restrict the adaptiveness of the organization to the goals of the administrative hierarchy” (page 36). Modern organizational research pushed forward this early theorizing, acknowledging that “knowledge transfer typically occurs across a boundary” and that “the boundary could be between occupational groups... between organizational units... or between geographic areas” (Argote and Miron-Spektor, 2011: 1131).

Despite its undoubted relevance for organizational effectiveness, boundary spanning across organizational subunits has proven to be a complex task. Organizations often mirror the “cavemen world” described by Watts (1999: 102), in which members live in dense isolated clusters (or “caves”) of strong, frequent, and redundant relations. Common membership in organizational subunits provides both frequent opportunities and stronger incentives to form within-subunit ties. Organizational subunits represent indeed social platforms (Pattison and Robins, 2002) that stimulate newcomers’ interaction (Morrison, 2002), facilitate interpersonal familiarity (Hinds et al., 2000), provide a repertoire of shared experiences (Marsden, 1988), and also generate the development of shared memory and elaboration of past events (March and Olsen, 1975). Therefore, sharing knowledge across subunit boundaries is more costly, in terms of time and effort to cultivate and maintain cross-boundary relations, than transferring knowledge within the subunit (e.g., Lomi and Zappa, 2015).

However, organizations are rarely composed of isolated caves: Ties within organizational subunits can represent the majority, but typically not all the ties among members (e.g., Zappa and Lomi, 2015). As Brass et al. (2004: 801) noted, “ties between people in different units are especially intriguing, because they create ties between organizational units.” So, what explains the likelihood to observe ties bridging

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